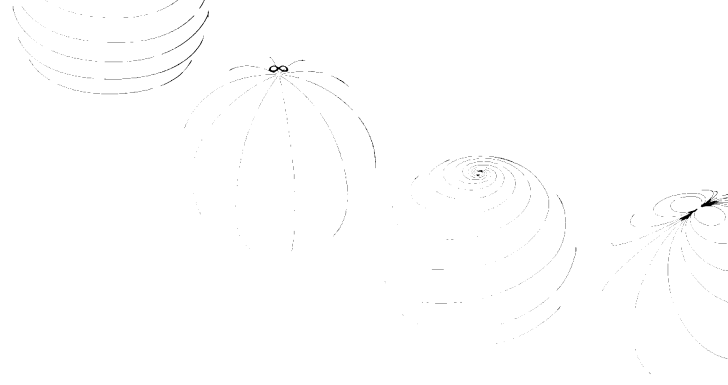




CHAPTER 9

Cauchy's Formula and Its Applications

9.1 Cauchy's Formula

9.1.1 Introduction

One of the principal objectives of this brief chapter is to tie up various loose ends from previous chapters. In particular, we have previously claimed (but have not yet explained) three important properties of an analytic function $f(z)$:

- We can differentiate $f(z)$ as many times as we please—it is “infinitely differentiable”.
- In the vicinity of an ordinary point, $f(z)$ can be expressed as a Taylor series.
- In the vicinity of a singularity, $f(z)$ can be expressed as a Laurent series.

The classical explanation¹ of these facts hinges on the following result. If $f(z)$ is analytic on and inside a simple loop L , and if a is a point inside L , then

$$\frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz = f(a). \quad (9.1)$$

This is called *Cauchy's Formula*—it constitutes the precise statement of the “rigidity” of analytic functions that we depicted in [5.3], p. 249. That is, the formula says that the values of f on L rigidly determine its values everywhere *inside* L .

We will give two explanations of (9.1), both of which are firmly rooted in Cauchy's Theorem.

¹ In the late 1950s a new approach was developed using topological ideas like those in Chapter 7, and it was our original intention to employ that approach here. However, having lacked both the time and the imagination to reduce the idea to its visual essentials, we have reluctantly fallen back on an integral-based approach. For more on the topological approach, see [Whyburn \(1955\)](#), [Whyburn \(2015\)](#) and [Beardon \(1979\)](#).

9.1.2 First Explanation

Since $f(z)$ is assumed analytic inside L , the function $[f(z)/(z - a)]$ is also analytic there, *except* that it has a single singularity at $z = a$. Thus it follows from (8.11), p. 454, that the value of the integral in (9.1) will not change if L is deformed into its interior without crossing a .

Let C_r be a circle of radius r , centred at a , and lying strictly inside L . Referring to [9.1a], we may deform L into such a circle without crossing a , and hence without altering the value of the integral:

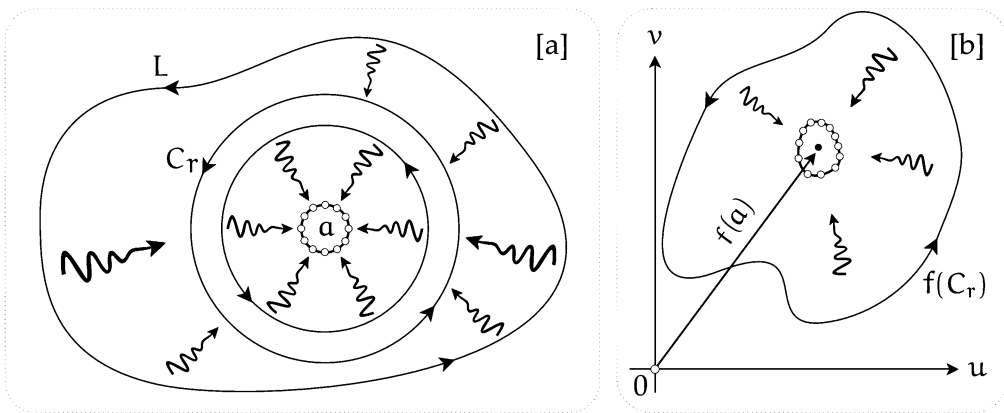
$$\frac{1}{2\pi i} \oint_L \frac{f(z)}{z - a} dz = \frac{1}{2\pi i} \oint_{C_r} \frac{f(z)}{z - a} dz. \quad (9.2)$$

The virtue of this transformation is that the integral round C_r turns out to have a simple and helpful interpretation.

First recall that the *average* value $\langle f \rangle_{C_r}$ of $f(z_\theta)$ as $z_\theta = a + r e^{i\theta}$ travels round the circle C_r is defined by

$$\langle f \rangle_{C_r} \equiv \frac{1}{2\pi} \int_0^{2\pi} f(z_\theta) d\theta.$$

In the previous chapter we saw geometrically that if θ increases by $d\theta$, causing z_θ to move dz along the circle, then $dz/(z - a) = i d\theta$. Substituting this into (9.2),



[9.1] First explanation of Cauchy's Formula. [a] If $f(z)$ is analytic inside L then so is $f(z)/(z - a)$, except at $z = a$, so the Deformation Theorem implies that we can deform L into the circle C_r of radius r without changing the value of the integral. But on C_r we can interpret the integral as the average value $\langle f \rangle_{C_r}$ of $f(z)$ on C_r , and, by the Deformation Theorem again, this average value is independent of the radius. If z_j are n equally spaced points around C_r then $\langle f \rangle_{C_r}$ can be visualized as the limit as $n \rightarrow \infty$ of the centroid of the image points $f(z_j)$. [b] As C_r shrinks towards a , these image points cluster more and more tightly around $f(a)$, so $\frac{1}{2\pi i} \oint_L \frac{f(z)}{z - a} dz = \langle f \rangle_{C_r} = f(a)$.

we find that the original integral round L may be interpreted as the average value of f on any of the circles C_r :

$$\frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz = \langle f \rangle_{C_r}.$$

Note in particular that $\langle f \rangle_{C_r}$ is independent of the radius r of the circle. To complete the derivation of (9.1), it therefore only remains to show that this radius-independent average is the value of f at the centre a .

To better grasp the meaning of the average value $\langle f \rangle_{C_r}$, imagine n equally spaced points $z_1, z_2, \dots, z_n$ on C_r , and let $w_1, w_2, \dots, w_n$ be their images under $z \mapsto w = f(z)$. The ordinary average $W_n \equiv \frac{1}{n} \sum_{j=1}^n w_j$ of these image points is their *centroid*, and $\langle f \rangle_{C_r}$ is the limiting position of W_n as n tends to infinity. [For a more detailed discussion of averages and centroids, consult the final section of Chapter 2.]

Now shrink the circle C_r towards its centre a , as illustrated in [9.1a]. Even if f is merely *continuous* (rather than analytic), $f(C_r)$ will shrink to $f(a)$, as indicated in [9.1b]. Since the images $w_1, w_2, \dots, w_n$ of any n points on C_r will all converge to $f(a)$, so will their centroid W_n . Thus

$$\lim_{r \rightarrow 0} \langle f \rangle_{C_r} = f(a),$$

and this completes our first explanation of Cauchy's Formula.

9.1.3 Gauss's Mean Value Theorem

In the course of the above investigation we have also picked up an interesting bonus result:

If $f(z)$ is analytic on and inside a circle C centred at a , then the average value of f on C is its value at the centre: $\langle f \rangle_C = f(a)$.

If we go on to split f into real and imaginary parts as $f = u + iv$, then we immediately deduce that $\langle u \rangle_C + i \langle v \rangle_C = u(a) + iv(a)$, and so

$$\langle u \rangle_C = u(a) \quad \text{and} \quad \langle v \rangle_C = v(a).$$

Thus if a real function Φ is either the real or the imaginary part of an analytic complex function, then its average on a circle is its value at the centre.

But if we are *given* a function Φ , how can we tell whether there exists an analytic function whose real or imaginary part is equal to Φ ? In Ex. 2, p. 293 you showed that a necessary condition is that Φ be *harmonic*, i.e., that it satisfy Laplace's equation,

$$\Delta \Phi \equiv (\partial_x^2 + \partial_y^2) \Phi = 0.$$

In fact in Chapter 12 we will see that this is also a *sufficient* condition, yielding Gauss's Mean Value Theorem:

The average value of a harmonic function on a circle is equal to the value of the function at the centre of the circle.

9.1.4 A Second Explanation and the General Cauchy Formula

What will happen to Cauchy's Formula if the loop L is *not* required to be simple? As in the previous chapter, it is now important to carefully define the "inside" of L as the set of points about which L has non-vanishing winding number:

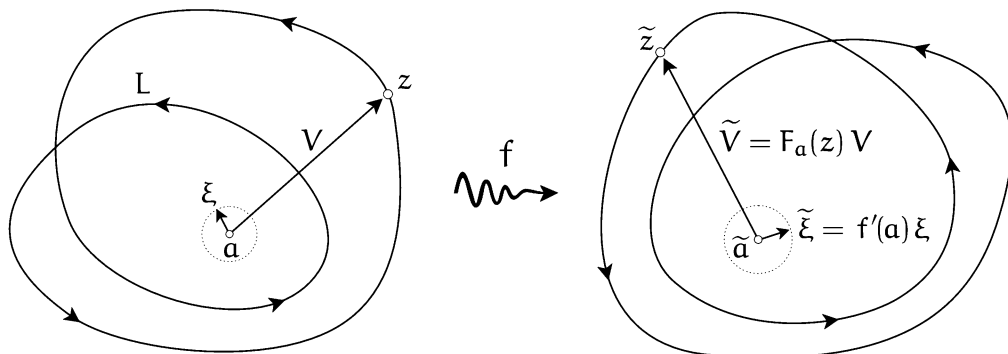
$$\text{"inside"} = \{p \mid v[L, p] \neq 0\}.$$

Suppose in [9.2] that f has no singularities "inside" L . Then the only interior singularity of $[f(z)/(z-a)]$ will be the one at $z = a$. Here, L winds round a twice, and it is clear that L may be deformed into a small circle centred at a and traversed $v[L, a] = 2$ times. By virtue of Cauchy's Formula for simple loops, we deduce that $\frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz = 2f(a)$.

More generally, this line of reasoning suggests the following *General Cauchy Formula*: If $f(z)$ is analytic on and "inside" a general loop L , then

$$\frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz = v[L, a] f(a). \quad (9.3)$$

That this is always true is not quite clear from the above line of reasoning. Certainly Hopf's Theorem [(7.2), p. 389] guarantees that without crossing the singularity at a , L may be deformed into a circle centred at a and traversed $v[L, a]$ times. But the singularities of f may be scattered in the midst of L , although (by assumption) none lie "inside" L . So is it clear that this deformation can always be performed without crossing any of these singularities?



[9.2] Second explanation and the General Cauchy Formula: $\frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz = v[L, a] f(a)$. Define $F_a(z)$ to be the non-infinitesimal analogue of the amplifactor, expanding and rotating V to its "image" $\tilde{V}$, so that $\tilde{V} = F_a(z)V$. Indeed, as V vanishes, the analogy becomes exact: $F_a(a) \equiv \lim_{z \rightarrow a} F_a(z) = f'(a)$. Since $F_a(z) = [f(z) - f(a)]/[z - a]$ is analytic inside L , its integral vanishes, by Cauchy's Theorem, and therefore $0 = \frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz - f(a) \frac{1}{2\pi i} \oint_L \frac{dz}{z-a} = \frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz - v[L, a] f(a)$, proving the result.

We encourage you to pursue this idea, but we shall now present a different approach which yields (9.3) cleanly and directly. Consider the mapping $z \mapsto \tilde{z} = f(z)$ in [9.2], and let us define

$$F_a(z) \equiv \frac{f(z) - f(a)}{z - a} = \frac{\tilde{z} - \tilde{a}}{z - a}.$$

If $V \equiv (z - a)$ is pictured as a vector emanating from a , and $\tilde{V} \equiv (\tilde{z} - \tilde{a})$ is pictured as its image emanating from $\tilde{a}$, then $F_a(z)$ describes the amount of rotation and expansion that carries V into $\tilde{V} = F_a(z) V$. Thus $F_a(z)$ is the non-infinitesimal analogue of the amplitwist $f'(a)$ that carries an infinitesimal vector ξ into its image $\tilde{\xi} = f'(a) \xi$, and

$$F_a(a) \equiv \lim_{z \rightarrow a} F_a(z) = f'(a).$$

Since $f(z)$ is assumed to be analytic and to have no singularities “inside” L , it follows that the same is true of $F_a(z)$. Thus the General Cauchy Theorem [(8.20), p. 472] implies that

$$\frac{1}{2\pi i} \oint_L F_a(z) dz = 0.$$

In other words,

$$\begin{aligned} 0 &= \frac{1}{2\pi i} \oint_L \frac{f(z)}{z - a} dz - f(a) \frac{1}{2\pi i} \oint_L \frac{dz}{z - a} \\ &= \frac{1}{2\pi i} \oint_L \frac{f(z)}{z - a} dz - \nu[L, a] f(a), \end{aligned}$$

as was to be shown.

9.2 Infinite Differentiability and Taylor Series

9.2.1 Infinite Differentiability

Returning to the case where L is a simple loop, let us show that if $f(z)$ is analytic inside L then so is $f'(z)$. From this it will follow by induction that $f(z)$ is *infinitely* differentiable.

What we must show is that if f is conformal, then so is f' . In other words, if f' is thought of as a mapping $z \mapsto \tilde{z} = f'(z)$, then each infinitesimal vector ξ emanating from a must be rotated and expanded *the same amount* to obtain the image vector $\tilde{\xi}$ emanating from $\tilde{a}$. That is, there is a single complex number $f''(a)$ (the amplitwist of f') such that $\tilde{\xi} = f''(a) \xi$.

Our first step is to obtain a neat expression for $f'(a)$ in terms of the values of $f(z)$ on L . Applying Cauchy's Formula to the analytic function $F_a(z)$, we deduce that

$$\begin{aligned} f'(a) = F_a(a) &= \frac{1}{2\pi i} \oint_L \frac{F_a(z)}{z-a} dz \\ &= \frac{1}{2\pi i} \oint_L \frac{f(z)}{(z-a)^2} dz - \frac{f(a)}{2\pi i} \oint_L \frac{dz}{(z-a)^2}. \end{aligned}$$

Since the second integral vanishes,

$$f'(a) = \frac{1}{2\pi i} \oint_L \frac{f(z)}{(z-a)^2} dz. \quad (9.4)$$

Now let's use this to find the image $\tilde{\xi}$ under $z \mapsto \tilde{z} = f'(z)$ of a short vector ξ emanating from a . Ignoring a term proportional to ξ^2 , we find [exercise] that

$$\tilde{\xi} \equiv f'(a + \xi) - f'(a) = \left[\frac{2}{2\pi i} \oint_L \frac{f(z)}{[z - (a + \xi)]^2 (z - a)} dz \right] \xi.$$

Allowing ξ to become infinitesimal, we deduce the desired result: every infinitesimal ξ emanating from a is amplitwisted to $\tilde{\xi} = f''(a) \xi$, where

$$f''(a) = \frac{2}{2\pi i} \oint_L \frac{f(z)}{(z-a)^3} dz. \quad (9.5)$$

Observe that since

$$\frac{d}{da} \left[\frac{1}{z-a} \right] = \frac{1}{(z-a)^2} \quad \text{and} \quad \frac{d^2}{da^2} \left[\frac{1}{z-a} \right] = \frac{2}{(z-a)^3},$$

both (9.4) and (9.5) are precisely what we would get if we simply differentiated the formula

$$f(a) = \frac{1}{2\pi i} \oint_L \frac{f(z)}{z-a} dz$$

with respect to a . Continuing in this way, we are led to conjecture that the n^{th} derivative $f^{(n)}$ may be represented as

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_L \frac{f(z)}{(z-a)^{n+1}} dz. \quad (9.6)$$

This is indeed true, as we shall see in a moment.

9.2.2 Taylor Series

Now let us show that if $f(z)$ is analytic on and inside an origin-centred circle C of radius R , then $f(z)$ may be expressed as a power series that converges inside this disc:

$$f(z) = c_0 + c_1 z + c_2 z^2 + c_3 z^3 + \dots$$

As we saw in Chapter 5, such a power series is infinitely differentiable within its disc of convergence. Thus the existence of the power series expansion will provide a second proof of the infinite differentiability of analytic functions. It also follows that the coefficients c_n may be expressed as

$$c_n = \frac{f^{(n)}(0)}{n!}, \quad (9.7)$$

so the power series is actually a Taylor series, and the coefficients do not depend on R :

$$f(z) = f(0) + f'(0)z + \frac{f''(0)}{2!}z^2 + \frac{f^{(3)}(0)}{3!}z^3 + \dots$$

To establish the existence of this series, we return to Cauchy's Formula (9.1). With a change of notation, this may be rewritten as

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(Z)}{Z-z} dZ = \frac{1}{2\pi i} \oint_C \frac{f(Z)}{Z} \left[\frac{1}{1-(z/Z)} \right] dZ.$$

See [9.3]. Since z is inside the circle on which Z lies, $|z| < |Z| = R$, and $|(z/Z)| < 1$. Thus $\frac{1}{1-(z/Z)}$ may be viewed as the sum of an infinite geometric series, and

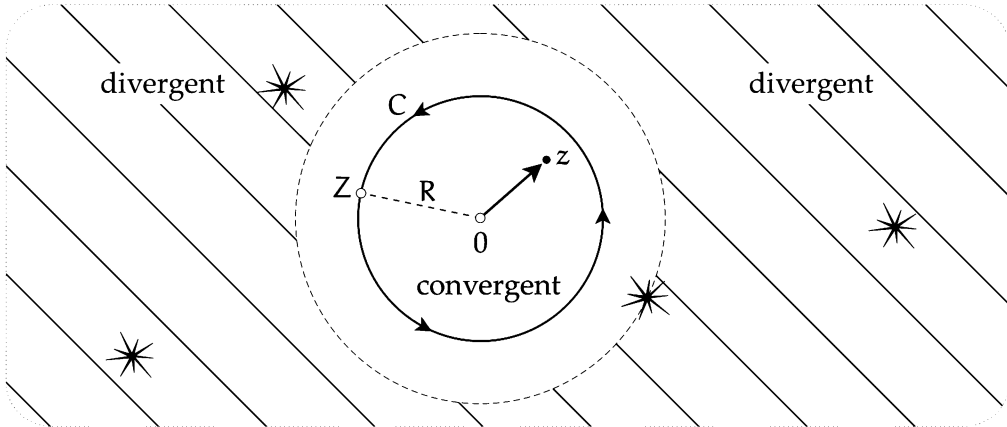
$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(Z)}{Z} [1 + (z/Z) + (z/Z)^2 + (z/Z)^3 + \dots] dZ.$$

Provided it makes sense to integrate this infinite series term by term, we deduce that $f(z)$ can indeed be expressed as a power series:

$$f(z) = \sum_{n=0}^{\infty} c_n z^n \quad \text{where} \quad c_n = \frac{1}{2\pi i} \oint_C \frac{f(Z)}{Z^{n+1}} dZ. \quad (9.8)$$

Furthermore, comparing this formula with (9.7), we also deduce (9.6).

To verify that this term-by-term integration is legitimate, consider the sum $f_N(z) \equiv \sum_{n=0}^{N-1} c_n z^n$ of the first N terms of the series (9.8). The result will be established if we can show that $f_N(z)$ tends to $f(z)$ as N tends to infinity.



[9.3] If $f(z)$ is analytic inside C , then it has a Taylor expansion there: $f(z) = \sum_{n=0}^{\infty} c_n z^n$, where $c_n = \frac{1}{2\pi i} \oint_C \frac{f(Z)}{Z^{n+1}} dZ$. Cauchy's Formula can be written as $f(z) = \frac{1}{2\pi i} \oint_C \frac{f(Z)}{Z} \left[\frac{1}{1-(z/Z)} \right] dZ$, and then $\left[\frac{1}{1-(z/Z)} \right]$ can be expanded as a geometric series. But the coefficients can also be obtained by differentiation, in the usual way, and therefore **we obtain a formula for the n^{th} derivative as an integral**: $f^{(n)}(a) = n! c_n = \frac{n!}{2\pi i} \oint_C \frac{f(Z)}{Z^{n+1}} dZ$.

Since

$$\frac{1}{1-(z/Z)} - [1 + (z/Z) + (z/Z)^2 + \cdots + (z/Z)^{N-1}] = \frac{(z/Z)^N}{1-(z/Z)},$$

it follows that

$$f(z) - f_N(z) = \frac{1}{2\pi i} \oint_C \frac{(z/Z)^N f(Z)}{(Z-z)} dZ.$$

Finally recall [see (8.5), p. 440] that the modulus of an integral cannot exceed the product of the length of the path and the maximum modulus of the integrand at points on the path. If M stands for the maximum value of $|f(Z)/(Z-z)|$ on C , then it follows that

$$|f(z) - f_N(z)| \leq RM|(z/Z)|^N.$$

Thus $\lim_{N \rightarrow \infty} f_N(z) = f(z)$, as was to be shown.

What is the radius of convergence of the series we have obtained? We know that if f is analytic inside C then the series (9.8) converges to $f(z)$ in that disc. Thus, referring to [9.3], C may be expanded up to the dashed circle, where it first encounters a singularity of f . More generally, $f(z)$ may be a single-valued branch of a multi-function, and as we learnt in Chapter 2, branch points then act as obstacles just as much as singularities. Thus *the radius of convergence is the distance from the centre of the expansion to the nearest singularity or branch point*.

One final point. We chose the origin as the centre of the expansion in order to avoid algebraic clutter, but this choice really involves no loss of generality. For suppose we instead choose the centre to be at a , meaning that we wish to expand $f(z)$ in powers of $\xi \equiv (z - a)$. If $f(z)$ is analytic at a then $F(\xi) \equiv f(a + \xi) = f(z)$ is analytic at the origin of the ξ -plane, and so it possesses an origin-centred Taylor expansion,

$$F(\xi) = \sum_{n=0}^{\infty} \frac{F^{(n)}(0)}{n!} \xi^n \implies f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n.$$

Alternatively [exercise], the existence of this series may be deduced by directly generalizing the argument leading to the origin-centred series. Either way, we conclude that

If $f(z)$ is analytic, and a is neither a singularity nor a branch point, then $f(z)$ may be expressed as the following power series, which converges to $f(z)$ within the disc whose radius is the distance from a to the nearest singularity or branch point:

$$f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n, \quad \text{where} \quad \frac{f^{(n)}(a)}{n!} = c_n = \frac{1}{2\pi i} \oint_L \frac{f(z)}{(z - a)^{n+1}} dz.$$

9.3 Calculus of Residues

9.3.1 Laurent Series Centred at a Pole

Suppose that a is a pole of an analytic function $f(z)$, i.e., $\lim_{z \rightarrow a} f(z) = \infty$. In Chapter 7 we investigated poles by assuming the existence of Taylor series (which we have just proven), and we found [see (7.19), p. 417] that near a we could express $f(z)$ as

$$f(z) = \frac{\phi(z)}{(z - a)^m},$$

where $\phi(z)$ is analytic, and $\phi(a) \neq 0$. Recall that the positive integer m is called the “order” of the pole, and that the greater the order of the pole, the faster $f(z)$ approaches ∞ as z approaches a .

We know that $\phi(z)$ can be expressed as a Taylor series centred at a :

$$\phi(z) = \sum_{n=0}^{\infty} c_n (z - a)^n, \quad \text{where} \quad c_n = \frac{\phi^{(n)}(a)}{n!}.$$

Hence we deduce that

If an analytic function $f(z)$ has a pole of order m at a , then in the vicinity of this pole, $f(z)$ possesses a Laurent series of the form

$$f(z) = \frac{c_0}{(z-a)^m} + \frac{c_1}{(z-a)^{m-1}} + \cdots + \frac{c_{m-1}}{(z-a)} + c_m + c_{m+1}(z-a) + \cdots.$$

Recall that the coefficient of $1/(z-a)$ is called the “residue” of $f(z)$ at a , denoted $\text{Res}[f, a]$. Also recall the crucial significance of the residue in evaluating integrals: if L is a simple loop containing a but no other singularities of f , then

$$\oint_L f(z) dz = 2\pi i \text{Res}[f, a].$$

More generally, suppose that L is not required to be simple, and that $f(z)$ has several poles, at a_1, a_2 , etc. The existence of the Laurent series was the missing ingredient in our discussion of this situation in the previous chapter. Having established that f does indeed possess a Laurent expansion in the vicinity of each of its poles, we have also verified the General Residue Theorem [(8.26), p. 477]:

$$\oint_L f(z) dz = 2\pi i \sum_n \nu[L, a_n] \text{Res}[f, a_n]. \quad (9.9)$$

9.3.2 A Formula for Calculating Residues

It is easy enough to find an explicit formula for the residue at a pole. Looking at the derivation of the Laurent series above, we see that

$$\text{Res}[f, a] = c_{m-1} = \frac{\phi^{(m-1)}(a)}{(m-1)!}.$$

Since $\phi(z) = (z-a)^m f(z)$, we deduce that

If a is an m^{th} order pole of $f(z)$, then

$$\text{Res}[f(z), a] = \frac{1}{(m-1)!} \left[\frac{d}{dz} \right]^{m-1} [(z-a)^m f(z)] \Big|_{z=a}. \quad (9.10)$$

From this general result one can derive other results that speed up the calculation of residues in commonly encountered special cases. For example, suppose that $f = (P/Q)$ has a “simple” (i.e., order 1) pole at a as a result of Q having a simple root at that point. In that case,

$$\text{Res}[f(z), a] = \lim_{z \rightarrow a} (z-a)f(z) = \lim_{z \rightarrow a} \frac{P(z)}{\left[\frac{Q(z)-Q(a)}{z-a} \right]}.$$

Thus, If $f(z) = \frac{P(z)}{Q(z)}$, and a is a simple root of Q , then

$$\text{Res}[f(z), a] = \frac{P(a)}{Q'(a)}. \quad (9.11)$$

For example, consider $f(z) = e^z/(z^4 - 1)$, which has simple poles at $z = \pm 1, \pm i$. If L is the circle $|z - 1| = 1$ then $z = 1$ is the only pole inside L , so (9.11) yields

$$\oint_L \frac{e^z}{z^4 - 1} dz = 2\pi i \operatorname{Res}[f, 1] = 2\pi i \left. \frac{e^z}{4z^3} \right|_{z=1} = \frac{1}{2}\pi i e.$$

We can actually check this using Cauchy's Formula. Since

$$(z^4 - 1) = (z - 1)(1 + z + z^2 + z^3),$$

we may write $f(z) = F(z)/(z - 1)$, where $F(z) \equiv e^z/(1 + z + z^2 + z^3)$. Since $F(z)$ is analytic inside L ,

$$\oint_L \frac{e^z}{z^4 - 1} dz = \oint_L \frac{F(z)}{z - 1} dz = 2\pi i F(1) = \frac{1}{2}\pi i e,$$

just as before.

9.3.3 Application to Real Integrals

In the exercises of the previous chapter we saw how certain kinds of real integrals could be expressed in terms of complex contour integrals. According to (9.9), the evaluation of contour integrals amounts to calculating residues, and we have just seen that this is straightforward. Thus the Residue Theorem leads to a powerful method of evaluating real integrals.

Historically, Cauchy's success in evaluating previously intractable real integrals was one of the first tangible signs of the power of his discoveries. Many modern texts (e.g., [Marsden et al. \(1999\)](#)) continue to celebrate this success with very detailed discussions of how the Residue Theorem may be applied to real integrals. However, there can be little doubt that this application is less important than it used to be. Today,² when faced with a tricky integral, a physicist, engineer, or mathematician is less likely to start calculating residues, and is more likely to reach for a computer. We will therefore only do a couple of illustrative examples, though further examples may be found in the exercises.

In Ex. 14, p. 480 we evaluated $\int_{-\infty}^{+\infty} (x^2 + 1)^{-2} dx$ using partial fractions. To redo this problem using residues, we integrate $f(z) = 1/(z^2 + 1)^2$ along the simple loop $(L + J)$ shown in [9.4a]. Here L is the segment of the real axis from $-R$ to $+R$, and J is the semi-circular contour (in the upper half plane) back from $+R$ to $-R$. Rewriting $f(z)$ as $f(z) = 1/(z + i)^2(z - i)^2$, we see that the only singularities are the second order poles at $z = \pm i$. Thus if $R > 1$ (as illustrated) then (9.10) yields

² Here, "today" refers to 1997! Today, in 2022, my claim is only *more* true than it was then.

$$\oint_{L+J} f(z) dz = 2\pi i \operatorname{Res}[f, i] = 2\pi i \left. \frac{d}{dz} \frac{1}{(z+i)^2} \right|_{z=i} = 2\pi i \frac{-2}{(2i)^3} = \frac{\pi}{2}.$$

But

$$\oint_{L+J} f(z) dz = \int_{-R}^{+R} \frac{dx}{(x^2+1)^2} + \int_J f(z) dz,$$

and, as you showed in the original exercise, the integral along J tends to zero as R tends to infinity. Thus

$$\int_{-\infty}^{+\infty} \frac{dx}{(x^2+1)^2} = \frac{\pi}{2}.$$

The famous physicist Richard Feynman once bet³ his colleagues, “I can do by other methods any integral anybody else needs contour integration to do.” It is a tribute to complex analysis that Feynman lost this bet. Nevertheless, we can check the above integral using a trick that frequently *did* enable Feynman to dispense with residues: differentiation of a simpler integral with respect to a parameter.

Consider the elementary result,

$$\int_{-\infty}^{+\infty} \frac{dx}{x^2+a^2} = \left[\frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) \right]_{-\infty}^{+\infty} = \frac{\pi}{a}.$$

Differentiating this with respect to a yields

$$\int_{-\infty}^{+\infty} \frac{2a}{(x^2+a^2)^2} dx = \frac{\pi}{a^2},$$

and substituting $a = 1$ then confirms our residue calculation.

For our second example, we will evaluate

$$I \equiv \int_0^{2\pi} \frac{d\theta}{\cos \theta + a}, \quad a > 1,$$

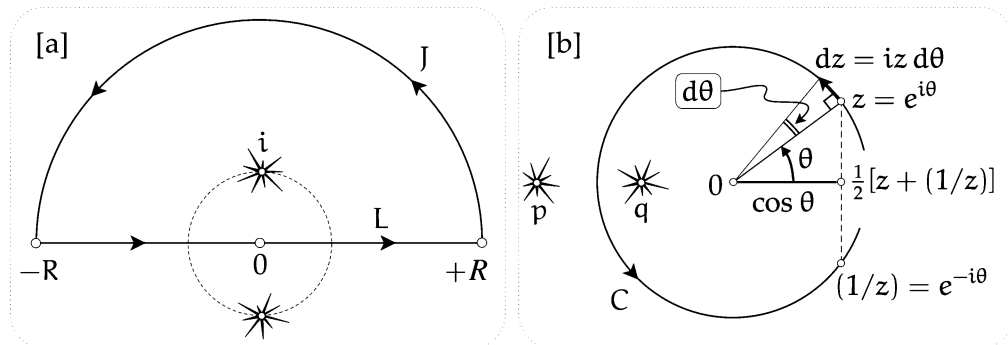
by rewriting it as a contour integral round the unit circle C . See [9.4b]. As illustrated, $\cos \theta$ is the midpoint of z and $(1/z)$, and dz is perpendicular to z and has length $d\theta$: in symbols, $\cos \theta = \frac{1}{2}[z + (1/z)]$ and $dz = iz d\theta$. Substituting into I ,

$$I = \oint_C \frac{(dz/iz)}{\frac{1}{2}[z + (1/z)] + a} = -2i \oint_C \frac{dz}{z^2 + 2az + 1}.$$

Since the singularities p and q of the integrand satisfy $pq = 1$, only one of them lies inside C —in fact p and q are geometric inverses. Thus [exercise],

$$I = 4\pi \operatorname{Res} \left[\frac{1}{(z-p)(z-q)}, q \right] = \frac{4\pi}{(q-p)} = \frac{2\pi}{\sqrt{a^2-1}}.$$

³ See Feynman (1997).



[9.4] Using residues to evaluate real integrals. **[a]** To evaluate $\int_{-\infty}^{+\infty} (x^2 + 1)^{-2} dx$, consider the integral of $f(z) = (z^2 + 1)^{-2}$. Then, $\oint_{L+J} f(z) dz = 2\pi i \operatorname{Res}[f, i] = \frac{\pi}{2}$. But as $R \rightarrow \infty$, $\oint_J f(z) dz \rightarrow 0$, and therefore $\int_{-\infty}^{+\infty} (x^2 + 1)^{-2} dx = \frac{\pi}{2}$. **[b]** To evaluate $I \equiv \int_0^{2\pi} \frac{d\theta}{\cos \theta + a}$, where $a > 1$, let $z = e^{i\theta}$, so that $\cos \theta = \frac{1}{2}[z + (1/z)]$ and $dz = iz d\theta$. Then $I = -2i \oint_C \frac{dz}{z^2 + 2az + 1}$. Factorizing the denominator into $(z-p)(z-q)$, we obtain $I = 4\pi \operatorname{Res}\left[\frac{1}{(z-p)(z-q)}, q\right] = \frac{4\pi}{(q-p)} = \frac{2\pi}{\sqrt{a^2-1}}$.

9.3.4 Calculating Residues using Taylor Series

In order to calculate a residue using (9.10), one must first know the order m of the pole. If $f(z)$ is built out of simple functions whose Taylor series are known, then the quickest method of finding m is by manipulating these series. Furthermore, this approach may be used to calculate the residue itself, often more easily than via formula (9.10). A few examples should suffice to explain the method.

For our first example, let $f(z) = (\sin^2 z / z^5)$, which clearly has a singularity of some kind at the origin. For small values of z , $\sin z \approx z$, so $f(z) \approx (1/z^3)$, and the order of the pole is therefore $m = 3$. By taking more terms of the Taylor series for $\sin z$ we can find more terms in the Laurent expansion of $f(z)$, and hence find the residue:

$$\begin{aligned} f(z) &= \frac{1}{z^5} \left[z - \frac{z^3}{6} + \cdots \right]^2 = \frac{1}{z^5} \left[z^2 - 2z \left(\frac{z^3}{6} \right) + \cdots \right] \\ &= \frac{1}{z^3} - \frac{1}{3z} + \cdots \\ \implies \operatorname{Res}[f, 0] &= -\frac{1}{3}. \end{aligned}$$

In order to appreciate how efficient this is, try checking the result using formula (9.10) instead.

Our next example will have valuable consequences. Let $g(z) = (1/z^2) \cot(\pi z)$, which is clearly singular at the origin. To find the order of this pole, and its residue, we begin by calculating the Laurent series of $\cot(\pi z)$. When doing such a calculation, it is important to remember that we are not trying to find the whole Laurent

series. We just want the $(1/z)$ term of g , which will come from the z term of $\cot \pi z$, so that's as far as we need go:

$$\begin{aligned}\cot \pi z &= \frac{\cos \pi z}{\sin \pi z} = \frac{\left[1 - \frac{(\pi z)^2}{2!} + \dots\right]}{\left[\pi z - \frac{(\pi z)^3}{3!} + \dots\right]} \\ &= \frac{1}{\pi z} \left[1 - \frac{(\pi z)^2}{2} + \dots\right] \left[1 - \frac{(\pi z)^2}{6} + \dots\right]^{-1} \\ &= \frac{1}{\pi z} \left[1 - \frac{(\pi z)^2}{2} + \dots\right] \left[1 + \frac{(\pi z)^2}{6} + \dots\right] \\ &= \frac{1}{\pi z} - \frac{\pi z}{3} + \dots\end{aligned}$$

In particular, note for future use that $\text{Res} [\cot(\pi z), 0] = (1/\pi)$.

Returning to the original function g , we find that

$$g(z) = \frac{1}{\pi z^3} - \frac{\pi}{3z} + \dots,$$

and so the origin is a triple pole with $\text{Res} [g, 0] = -(\pi/3)$. Again, try checking this using formula (9.10) instead.

Continuing with this example, it's clear that $g(z)$ also has a singularity at each integer n . To find the residue at n , we could write $z = n + \xi$ and expand g as a Laurent series in powers of ξ . However, this is unnecessary. Since $(1/z^2)$ is non-singular at n , and since $\cot[\pi(n + \xi)] = \cot \pi \xi$,

$$\begin{aligned}\text{Res} [(1/z^2) \cot(\pi z), n] &= (1/n^2) \text{Res} [\cot(\pi z), n] \\ &= (1/n^2) \text{Res} [\cot(\pi z), 0] \\ &= 1/(\pi n^2).\end{aligned}$$

More generally, note that if $f(z)$ is any analytic function that is non-singular at n , then

$$\text{Res} [f(z) \cot(\pi z), n] = \frac{1}{\pi} f(n). \quad (9.12)$$

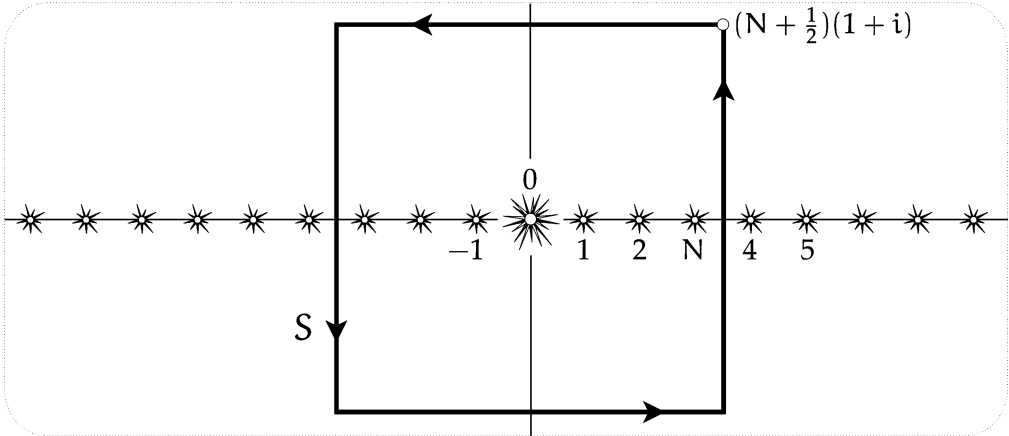
This may also be verified [exercise] using (9.11).

9.3.5 Application to Summation of Series

Historically, $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$ was the first simple-looking series that mathematicians were unable to sum using elementary algebraic methods. After the Bernoulli family had tried and failed, Euler finally cracked the problem in 1734 by means of a brilliantly unorthodox argument⁴. The answer he found was as unexpected as his methods:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

⁴ See Ex. 13 and Stillwell (2010).



[9.5] Evaluation of $\sum_{n=1}^{\infty} f(n)$ using $\text{Res}[f(z)\cot(\pi z), n] = \frac{1}{\pi}f(n)$. If we take $f(z) = 1/z^2$, then $\text{Res}\left[\frac{\cot(\pi z)}{z^2}, n\right] = \frac{1}{\pi n^2}$. Also, one finds [see text] that $\text{Res}\left[\frac{\cot(\pi z)}{z^2}, 0\right] = -\frac{\pi}{3}$. Therefore, $\frac{1}{2\pi i} \oint_S \frac{\cot(\pi z)}{z^2} dz = -\frac{\pi}{3} + \frac{2}{\pi} \sum_{n=1}^N \frac{1}{n^2}$. But as $N \rightarrow \infty$, one finds [see text] that this integral goes to zero, and we thereby obtain Euler's extraordinary discovery of 1734: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Today such results can be derived in a systematic way using residues. Reconsider the function $g(z) = (1/z^2)\cot(\pi z)$ above. With N a positive integer, let S be the origin-centred square with vertices $(N + \frac{1}{2})(\pm 1 \pm i)$ shown in [9.5]. Adding up the residues of the illustrated singularities inside S ,

$$\begin{aligned} \frac{1}{2\pi i} \oint_S g(z) dz &= \text{Res}[g(z), 0] + \sum_{n=-N}^{-1} \text{Res}[g(z), n] + \sum_{n=1}^N \text{Res}[g(z), n] \\ &= -\frac{\pi}{3} + \frac{2}{\pi} \sum_{n=1}^N \frac{1}{n^2}. \end{aligned}$$

As we will now see, the integral on the LHS tends to zero as N tends to infinity, and from this fact we immediately deduce Euler's result.

To show that the integral of $g(z) = (1/z^2)\cot(\pi z)$ does indeed tend to zero as S expands, we must show that the size of the integrand dies away faster than the perimeter ($8N + 4$) of S grows. First the easy part: $|g(z)| = |1/z^2| \cdot |\cot(\pi z)|$, and on S we clearly have $|z| > N$, so $|1/z^2| < (1/N^2)$.

Next we must examine the size of

$$|\cot(\pi z)| = \left| \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} \right|$$

on the four edges of S . We begin with the horizontal edges, $y = \pm(N + \frac{1}{2})$. Since $|e^{\pm i\pi z}| = e^{\mp \pi y}$, it is not hard to see [exercise] that if N is reasonably large then $|\cot(\pi z)|$ is very close to 1. Thus for sufficiently large N , $|\cot(\pi z)|$ will certainly be less than 2, for example.

Finally, on the vertical edges we have $z = \pm(N + \frac{1}{2}) + iy$, and it follows [exercise] that

$$|\cot(\pi z)| = \left| \frac{1 - e^{-2\pi y}}{1 + e^{-2\pi y}} \right| \leq 1.$$

For sufficiently large N , we have established that $|\cot(\pi z)| < 2$ everywhere on S , so by virtue of (8.5), p. 440,

$$\left| \oint_S g(z) dz \right| \leq (\text{Max } |g| \text{ on } S)(\text{perimeter of } S) < \frac{2}{N^2}(8N + 4).$$

Since the RHS tends to zero as N tends to infinity, we are done.

More generally, let $f(z)$ be an analytic function such that $|f(z)| < (\text{const.})/|z|^2$ for sufficiently large $|z|$. Then it is clear that the above argument applies equally well to the integral of $f(z) \cot(\pi z)$:

$$\begin{aligned} 0 &= \lim_{N \rightarrow \infty} \frac{1}{2\pi i} \oint_S f(z) \cot(\pi z) dz \\ &= \sum_{\text{all poles}} \text{Res} [f(z) \cot(\pi z)] \\ &= \sum_{n=-\infty}^{\infty} \text{Res} [f(z) \cot(\pi z), n] + \sum_{\text{poles of } f(z)} \text{Res} [f(z) \cot(\pi z)] \\ &= \frac{1}{\pi} \sum_{n=-\infty}^{\infty} f(n) + \sum_{\text{poles of } f(z)} \text{Res} [f(z) \cot(\pi z)], \end{aligned}$$

where the last equality follows from (9.12).

Thus

If $f(z)$ is an analytic function such that $|f(z)| < (\text{const.})/|z|^2$ for sufficiently large $|z|$, then

$$\sum_{n=-\infty}^{\infty} f(n) = -\pi \sum_{\text{poles of } f(z)} \text{Res} [f(z) \cot(\pi z)]. \quad (9.13)$$

Of course if any of the poles of $f(z)$ happen to be integers, then these values of n are understood to be excluded from the LHS of (9.13).

Note that while symmetry enables us to calculate sums like $\sum_{n=1}^{\infty} (1/n^2)$ and $\sum_{n=1}^{\infty} (1/n^4)$ using (9.13), we *cannot* use (9.13) to calculate a sum like $\sum_{n=1}^{\infty} (1/n^3)$. What, you might ask, is the sum of this last series? The answer is that *nobody knows!*⁵

⁵ This was written in 1997, but, 25 years later, it *remains* a mystery.

As a further interesting illustration of (9.13), consider $f(z) = 1/(z - w)^2$, where w is an arbitrary (non-integer) complex number. Geometrically, $|z - w|$ is the distance w to z , and this makes it easy to see that $|f(z)|$ satisfies the requirement of the theorem. Since the only singularity of $f(z)$ is a double pole at $z = w$,

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n - w)^2} = -\pi \operatorname{Res} \left[\frac{\cot(\pi z)}{(z - w)^2}, w \right].$$

Using formula (9.10),

$$\operatorname{Res} \left[\frac{\cot(\pi z)}{(z - w)^2}, w \right] = \frac{d}{dz} \cot(\pi z) \Big|_{z=w} = -\frac{\pi}{\sin^2(\pi w)}.$$

Thus we obtain the remarkable result,

$$\frac{\pi^2}{\sin^2(\pi w)} = \cdots + \frac{1}{(2 + w)^2} + \frac{1}{(1 + w)^2} + \frac{1}{w^2} + \frac{1}{(1 - w)^2} + \frac{1}{(2 - w)^2} + \cdots.$$

Such series were first discovered by Euler in 1748. What is remarkable about such a formula is that the periodicity of the function on the LHS is *explicitly* exhibited by the series on the RHS. That is, if you change w to $(w + 1)$, the series is clearly unaltered.

9.4 Annular Laurent Series

9.4.1 An Example

We have seen that the Laurent series is the natural generalization of the Taylor series when the centre of the expansion is a pole rather than a non-singular point. However, this is by no means the *only* situation in which Laurent series are needed.

For example, consider

$$F(z) = \frac{1}{(1 - z)(2 - z)},$$

whose simple poles are illustrated in [9.6a]. Since F is analytic within the unit disc, it possesses a Taylor series in powers of z . This may be found most easily by splitting F into partial fractions:

$$\begin{aligned} F(z) &= \frac{1}{(1 - z)} - \frac{1}{(2 - z)} \\ &= \frac{1}{(1 - z)} - \frac{1}{2[1 - (z/2)]} = \overbrace{\sum_{n=0}^{\infty} z^n}^{\text{for } |z| < 1} - \frac{1}{2} \overbrace{\sum_{n=0}^{\infty} (z/2)^n}^{\text{for } |z| < 2} \\ &= \frac{1}{2} + \frac{3}{4}z + \frac{7}{8}z^2 + \cdots + [1 - (1/2)^{n+1}] z^n + \cdots, \quad \text{for } |z| < 1. \end{aligned}$$

The pole at $z = 1$ means that outside the unit disc F cannot be expressed as a power series in z . However, in the shaded annulus $1 < |z| < 2$ it can be expressed as a *Laurent* series in z :

$$\begin{aligned} F(z) &= -\frac{1}{z[1-(1/z)]} - \frac{1}{2[1-(z/2)]} = -\overbrace{\sum_{n=0}^{\infty} (1/z)^{n+1}}^{\text{for } |z|>1} - \frac{1}{2} \overbrace{\sum_{n=0}^{\infty} (z/2)^n}^{\text{for } |z|<2} \\ &= \cdots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - \frac{1}{2} - \frac{z}{4} - \frac{z^2}{8} - \frac{z^3}{16} - \cdots, \quad \text{for } 1 < |z| < 2. \end{aligned}$$

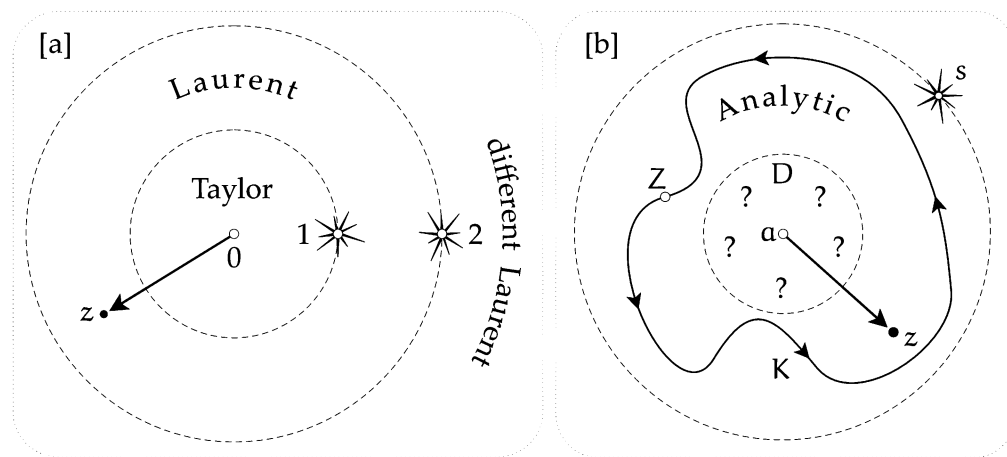
Finally, in the region $|z| > 2$ beyond the annulus we obtain [exercise] a *different* Laurent series:

$$F(z) = \frac{1}{z^2} + \frac{3}{z^3} + \cdots + \frac{(2^{n-1} - 1)}{z^n} + \cdots, \quad \text{for } |z| > 2.$$

9.4.2 Laurent's Theorem

What we have just seen is an illustration of a general phenomenon. See [9.6b].

If $f(z)$ is analytic everywhere within an annulus A centred at a , then $f(z)$ can be expressed as a Laurent series within A . In fact, if K is any simple loop lying within A and winding once round a ,



[9.6] Laurent Series. [a] Let $F(z) = \frac{1}{(1-z)(2-z)}$. Then $F(z)$ has a Taylor series within the unit disc, but the singularity at $z = 1$ means that no power series can exist beyond. However, in the shaded annulus, $1 < |z| < 2$, it is possible to express $F(z)$ as a so-called **Laurent series that has positive and negative powers of z** : $F(z) = \cdots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - \frac{1}{2} - \frac{z}{4} - \frac{z^2}{8} - \frac{z^3}{16} - \cdots$. Moving outside the annulus, a Taylor series is again impossible because of the two poles closer in, but, here too, a Laurent series exists. However, it is now a different Laurent series: $F(z) = \frac{1}{z^2} + \frac{3}{z^3} + \cdots + \frac{(2^{n-1}-1)}{z^n} + \cdots$. [b] **Laurent's Theorem.** If $f(z)$ is analytic everywhere within the annulus, then $f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n$, where $c_n = \frac{1}{2\pi i} \oint_K \frac{f(Z)}{(Z-a)^{n+1}} dZ$.

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n, \quad \text{where} \quad c_n = \frac{1}{2\pi i} \oint_K \frac{f(Z)}{(Z-a)^{n+1}} dZ. \quad (9.14)$$

Before establishing this result, which is called *Laurent's Theorem*, we make the following observations regarding its significance:

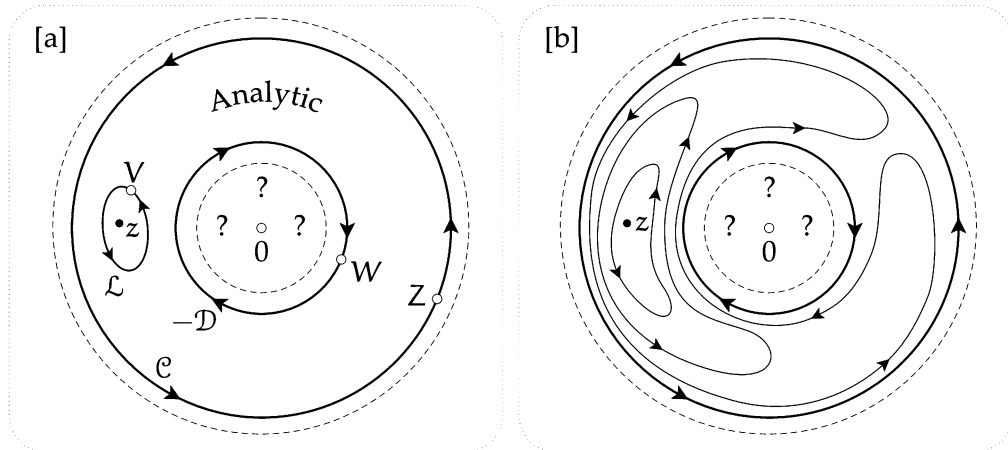
- The surprising thing about the result is the *existence* of a Laurent series, not the fact that it converges in an annulus. Since we know that a power series in $(z-a)$ will converge inside a disc centred at a , it follows [exercise] that a power series in $1/(z-a)$ will converge *outside* a disc centred at a . Since a Laurent series is (by definition) the sum of a power series in $(z-a)$ and a power series in $1/(z-a)$, it follows that it will converge in an annulus.
- Previously we were able to deduce the existence of a Laurent series only in the vicinity of a pole. The present result is much more powerful: as indicated by the question marks in [9.6b], we make no assumptions at all concerning the behaviour of $f(z)$ in the disc D bounded by the inner edge of the annulus. In practice, the outer edge of the annulus may be expanded until it hits a singularity s of $f(z)$, and the inner edge may likewise be contracted until it hits the outermost singularity lying in D .
- If there are *no* singularities in D , then the inner edge of the annulus may be completely collapsed, thereby transforming the annulus into a disc. In this case, (9.14) does not contain any negative powers. For if n is negative then $f(z)/(z-a)^{n+1}$ is analytic everywhere inside K , and so $c_n = 0$. In this way we recover the existence of Taylor's series as a special case of Laurent's Theorem.
- Suppose that a is a singularity and that for a sufficiently small value of ϵ there are no other singularities within a distance ϵ of a . In this case one says that a is an *isolated singularity* of $f(z)$. Applying Laurent's Theorem to the annulus $0 < |z-a| < \epsilon$, we find that there are just two fundamentally different possibilities: the principal part of the Laurent series either has finitely many terms, or infinitely many terms. Recall that in the latter case we have (by definition) an "essential singularity". See p. 417, where we considered the example

$$e^{1/z} = 1 + \frac{1}{1!z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \cdots.$$

To sum up,

An isolated singularity of an analytic function is either a pole or an essential singularity.

Now let us establish (9.14). In order to simplify the calculations, we will only treat the case $a = 0$, illustrated in [9.7a]. Here, z is a general point in the annulus, $\mathcal{C}$ and $\mathcal{D}$ are counterclockwise circles such that z lies between them, and $\mathcal{L}$ is a simple loop round z , lying within the annulus.



[9.7] Proof of Laurent's Theorem. [a] The initial key step mimics the rewriting of Cauchy's Formula that was used to prove Taylor's Theorem: $f(z) = \frac{1}{2\pi i} \oint_{\mathcal{L}} \frac{f(V)}{V-z} dV = \frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{f(Z)}{Z} \left[\frac{1}{1-(z/Z)} \right] dZ + \frac{1}{2\pi i} \oint_{\mathcal{D}} \frac{f(W)}{z} \left[\frac{1}{1-(W/z)} \right] dW$. The theorem then follows quickly by expanding both square brackets into geometric series, in z and $(1/z)$, respectively. [b] This picture justifies the previous equation by showing that $\mathcal{L}$ can indeed be deformed within the annulus into $(\mathcal{C}) + (-\mathcal{D})$, without changing the value of the integral.

First, by Cauchy's Formula,

$$f(z) = \frac{1}{2\pi i} \oint_{\mathcal{L}} \frac{f(V)}{V-z} dV = \frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{f(Z)}{Z-z} dZ - \frac{1}{2\pi i} \oint_{\mathcal{D}} \frac{f(W)}{W-z} dW,$$

where the second equality follows from the fact that $\mathcal{L}$ may be deformed within the annulus into $(\mathcal{C}) + (-\mathcal{D})$, as indicated in [9.7b].

Next, we rewrite the above equation as

$$f(z) = \frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{f(Z)}{Z} \left[\frac{1}{1-(z/Z)} \right] dZ + \frac{1}{2\pi i} \oint_{\mathcal{D}} \frac{f(W)}{z} \left[\frac{1}{1-(W/z)} \right] dW.$$

The significance of this is that $|z/Z| < 1$ and $|W/z| < 1$, so both integrands on the RHS can be expanded into geometric series, very much as we did in the example of [9.6a].

Referring back to the derivation of the Taylor series (9.8), the integral round $\mathcal{C}$ can be expressed as

$$\frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{f(Z)}{Z} \left[\frac{1}{1-(z/Z)} \right] dZ = \sum_{n=0}^{\infty} \left[\frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{f(Z)}{Z^{n+1}} dZ \right] z^n.$$

Essentially identical reasoning [exercise] also justifies term-by-term integration in the case of the integral round $\mathcal{D}$:

$$\frac{1}{2\pi i} \oint_{\mathcal{D}} \frac{f(W)}{z} \left[\frac{1}{1 - (W/z)} \right] dW = \sum_{n=1}^{\infty} \left[\frac{1}{2\pi i} \oint_{\mathcal{D}} W^{n-1} f(W) dW \right] \left(\frac{1}{z} \right)^n.$$

Thus the existence of the Laurent series is established:

$$f(z) = \cdots + \frac{d_3}{z^3} + \frac{d_2}{z^2} + \frac{d_1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots,$$

where

$$d_m = \frac{1}{2\pi i} \oint_{\mathcal{D}} W^{m-1} f(W) dW \quad \text{and} \quad c_n = \frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{f(Z)}{Z^{n+1}} dZ.$$

Finally, the following two observations enable us to tidy up the result. First, by the Deformation Theorem [p. 454], the integrals defining d_m and c_n do not change their values if we allow $\mathcal{C}$ to contract and $\mathcal{D}$ to expand till they coalesce into the *same* circle. Indeed, we may replace both $\mathcal{C}$ and $\mathcal{D}$ with any simple loop K contained in the annulus and winding round it once. Second, if we write $m = -n$ then the integral defining the coefficient d_{-n} of z^n has integrand $W^{-n-1} f(W) = f(W)/W^{n+1}$, which is the same as the integrand for the c_n 's. Thus, as was to be shown, the Laurent series may be expressed in the compact form of (9.14):

$$f(z) = \sum_{n=-\infty}^{\infty} c_n z^n, \quad \text{where} \quad c_n = \frac{1}{2\pi i} \oint_K \frac{f(Z)}{Z^{n+1}} dZ.$$

9.5 Exercises

- 1 If C is the unit circle, show that

$$\int_0^{2\pi} \frac{dt}{1 + a^2 - 2a \cos t} = \oint_C \frac{i dz}{(z - a)(az - 1)}.$$

Use Cauchy's Formula to deduce that if $0 < a < 1$, then

$$\int_0^{2\pi} \frac{dt}{1 + a^2 - 2a \cos t} = \frac{2\pi}{1 - a^2}.$$

- 2 Let $f(z)$ be analytic on and inside a circle K defined by $|z - a| = \rho$, and let M be the maximum of $|f(z)|$ on K .

(i) Use (9.6) to show that

$$|f^{(n)}(a)| \leq \frac{n!M}{\rho^n}.$$

(ii) Suppose that $|f(z)| \leq M$ for all z , where M is some constant. By putting $n = 1$ in the above inequality, rederive Liouville's Theorem [p. 410].

(iii) Suppose $|f(z)| \leq M|z|^n$ for all z , where n is some positive integer. Show that $f^{(n+1)}(z) \equiv 0$, and deduce that $f(z)$ must be a polynomial whose degree does not exceed n .

- 3 (i) Show that if C is any simple loop round the origin, then

$$\binom{n}{r} = \frac{1}{2\pi i} \oint_C \frac{(1+z)^n}{z^{r+1}} dz.$$

(ii) By taking C to be the unit circle, deduce that

$$\binom{2n}{n} \leq 4^n.$$

For other interesting applications of complex analysis to problems involving binomial coefficients, see [Bak and Newman \(2010\)](#).

- 4 The *Legendre polynomials* $P_n(z)$ are defined by

$$P_n(z) = \frac{1}{2^n n!} \frac{d^n}{dz^n} [(z^2 - 1)^n].$$

These polynomials are important in many physical problems, including the quantum mechanical description of the hydrogen atom.

(i) Calculate $P_1(z)$ and $P_2(z)$, and explain why P_n has degree n .

(ii) Use (9.6) to show that

$$P_n(z) = \frac{1}{2\pi i} \oint_K \frac{(Z^2 - 1)^n}{2^n (Z - z)^{n+1}} dZ,$$

where K is any simple loop round z .

(iii) By taking K to be a circle of radius $\sqrt{|z^2 - 1|}$ centred at z , deduce that

$$P_n(z) = \frac{1}{\pi} \int_0^\pi (z + \sqrt{z^2 - 1} \cos \theta)^n d\theta.$$

(iv) Check that this last formula yields the same $P_1(z)$ and $P_2(z)$ as you obtained in part (i).

5 If C denotes the unit circle, show that

$$\int_0^{2\pi} \frac{\sin^2 \theta}{5 - 4 \cos \theta} d\theta = -\frac{i}{4} \oint_C \frac{(z^2 - 1)^2}{z^2(z - 2)(2z - 1)} dz = \frac{\pi}{4}.$$

6 Let $f(z)$ be an analytic function with no poles on the real axis, and such that $|f(z)| < (\text{const.})/|z|^2$ for sufficiently large $|z|$. By integrating $f(z) e^{iz}$ along the contour $(L + J)$ shown in [9.4a], deduce that

$$\int_{-\infty}^{+\infty} f(x) \cos x \, dx + i \int_{-\infty}^{+\infty} f(x) \sin x \, dx = 2\pi i \sum_{\text{upper half-plane}} \text{Res} [f(z) e^{iz}].$$

[Hint: First show that if $y > 0$, then $|e^{iz}| < 1$.]

7 Use the result of the previous exercise to do the following problems, in which we assume that $a > 0$.

(i) Show that

$$\int_{-\infty}^{+\infty} \frac{\cos x}{x^2 + a^2} dx = \frac{\pi}{a} e^{-a}.$$

(ii) Evaluate

$$\int_{-\infty}^{+\infty} \frac{x \sin x}{(x^2 + a^2)^2} dx.$$

8 Let $F_n(z) = 1/(1 + z^n)$, where n is even.

(i) Use (9.11) to show that if p is a pole of F_n then $\text{Res} [F_n, p] = -(p/n)$.

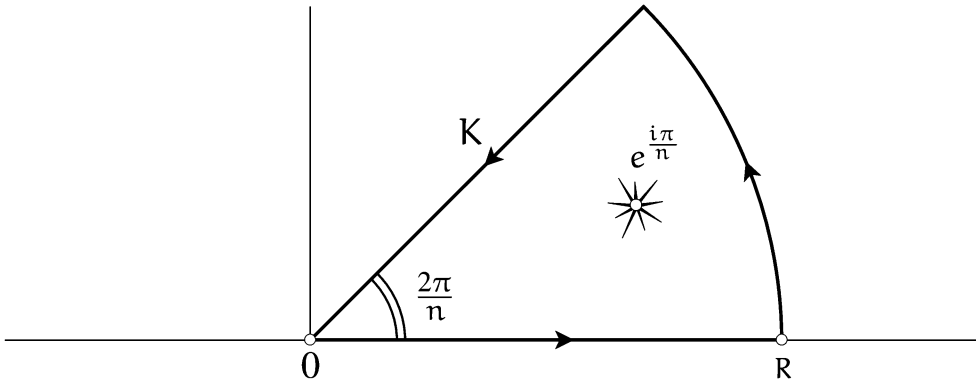
(ii) With the help of part (i), show that the sum of the residues of F_n in the upper half-plane is a geometric series with sum $1/[in \sin(\pi/n)]$.

(iii) By applying the Residue Theorem to the contour $(L + J)$ shown in [9.4a], deduce that

$$\int_0^\infty \frac{dx}{1 + x^n} = \frac{\pi}{n \sin(\pi/n)}. \quad (9.15)$$

(iv) Although the above derivation breaks down when n is odd [why?], use a computer to verify that (9.15) is nevertheless still true.

9 Continuing from the previous question, consider the wedge-shaped contour K shown below.



- (i) Use the Residue Theorem to show that if $n = 2, 3, 4, \dots$, and $R > 1$ (as illustrated), then

$$\oint_K \frac{dz}{1+z^n} = -\frac{2\pi i}{n} e^{i(\pi/n)}.$$

- (ii) Show that

$$\lim_{R \rightarrow \infty} \oint_K \frac{dz}{1+z^n} = \left[1 - e^{i(2\pi/n)} \right] \int_0^\infty \frac{dx}{1+x^n}.$$

- (iii) Deduce that (9.15) is indeed valid for odd n as well as even n .

- 10 Use (9.13) to show that $\sum_{n=1}^\infty (1/n^4) = (\pi^4/90)$.

- 11 Show that if $f(z)$ is an analytic function such that $|f(z)| < (\text{const.})/|z|^2$ for sufficiently large $|z|$, then

$$\sum_{n=-\infty}^\infty (-1)^n f(n) = -\pi \sum_{\text{poles of } f(z)} \text{Res} [f(z) \operatorname{cosec}(\pi z)].$$

In this formula, it is understood that if any of the poles of $f(z)$ happen to be integers, then these values of n are excluded from the LHS.

- 12 Use the result of the previous question to do the following:

- (i) Show that

$$1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots = \frac{\pi^2}{12}.$$

- (ii) Find the sum of the series

$$\frac{1}{2} - \frac{1}{5} + \frac{1}{10} - \frac{1}{17} + \dots + \frac{(-1)^{n+1}}{(n^2+1)} + \dots$$

- 13 (i) Show that

$$\sum_{n=-\infty}^\infty \frac{1}{z^2 - n^2} = \frac{\pi \cot \pi z}{z}.$$

- (ii) Show that the previous equation can be rewritten as

$$\cot z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2\pi^2}.$$

- (iii) Show that the previous equation can be rewritten as

$$\frac{d}{dz} [\ln(\sin z/z)] = \sum_{n=1}^{\infty} \frac{d}{dz} \ln(z^2 - n^2\pi^2).$$

- (iv) By integrating along any path from 0 to z that avoids integers, and then exponentiating both sides of the resulting equation, deduce that

$$\sin z = z \left(1 - \frac{z^2}{\pi^2}\right) \left(1 - \frac{z^2}{2^2\pi^2}\right) \left(1 - \frac{z^2}{3^2\pi^2}\right) \cdots.$$

[Hint: Recall that $\lim_{z \rightarrow 0} (\sin z/z) = 1$.]

This famous formula was discovered by Euler in 1734, in a much simpler way, via an extraordinary leap of imagination; see [Stillwell \(2010\)](#) for the details.

- (vi) Multiplying out the brackets on the right, we see that the z^3 terms arise from taking a z^2 term from a single bracket, and 1's from all the others. But, on the left hand side, the power series for $\sin z$ is known. Thus,

$$z - \frac{1}{3!}z^3 + \cdots = z - \left[\frac{1}{\pi^2} + \frac{1}{2^2\pi^2} + \frac{1}{3^2\pi^2} + \frac{1}{4^2\pi^2} + \cdots \right] z^3 + \cdots.$$

Euler was thereby able to sum a series that had baffled the greatest mathematicians that had come before him, making the wonderful and startling discovery that

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6}.$$

